

EVAPORATION

Final Lab Report 1

Commented [DL1]: 89/100

Unit Operations CHE 382

February 05, 2020

Group 1, Thursday, AM section

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1. Executive Summary

The objective of this experiment is to study the effects of pressure change or vacuum levels on the concentrated solution which consists of a nonvolatile solute (sugar) and a volatile solvent (water). Parameters such as solvent volume, economy, and capacity will be recorded by considering the effects of pressure and temperature. Furthermore, the effect of evaporation on liquid characteristics such as concentration, foaming, scale and temperature sensitivity will be considered throughout the experiment.

As shown in the results section the data showed a linearly increasing trend for both the capacity vs. vacuum pressure and economy vs. vacuum pressure. Initially at 10 inches Hg vacuum pressure, it has an economy of 0.238 and capacity of 1.784 kg/h. And for 20 in Hg an economy of 0.411 and 3.92 kg/h capacity was calculated. This gives the confirmation of increasing vacuum pressure correlating to an increase in both economy and capacity from the evaporator unit. The same should be said about the heat transfer coefficient, but in the results, there is a decrease from 10 to 15 inches Hg and then an increase from 15 to 20 inches Hg. The heat transfer coefficient at 10 inches Hg 2121 W/m²·K to a decrease of about 40 W/m²·K, 2081 W/m²·K, at 15 inches Hg. From there it increased by about 60 W/m²·K, 2141 W/m²·K, at 20 inches Hg. There should have been a constant increase in heat transfer coefficient as the vacuum pressure increased.

Commented [DL2]: 9/10

Commented [DL3]: How much?

Commented [DL4]: Better to use SI unit.

Commented [DL5]: The 15 in. Hg data?

Commented [DL6]: Good.

2. Introduction

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Evaporation is an equilibrium-based separation process by which a solution consisting of a non-volatile solute and a volatile solvent mostly water brought to a concentration. The product of evaporation is either a heavy liquid or liquor and a vapor consisting of a portion of the solvent. The vapor is usually condensed and removed. Evaporation is distinct from drying, crystallization and distillation as a process in several important ways. Evaporation produces a liquid product (sometimes a very viscous one) which separates it from drying that in turn produces a solid. It puts no emphasis on forming crystals which clearly separates it from crystallization. Finally, it differs from distillation in that the vapor produced is often single component. However, there is no emphasis on or specific step for separation of the vapor mixture if it is a mixture. The objective of this lab is to concentrate a solution which consists of a nonvolatile solute (sugar) and a volatile solvent (water).

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This experiment was performed to evaluate the performance of a single effect evaporator under different vacuum pressure by calculating the capacity and economy at these pressures. The best pressure for the system can be figured out which is important to minimize the energy loss in the system and increase the performance of the evaporator.

Commented [DL9]: Good.

The single effect evaporator is consisted of a heat exchanger, steam trap, condenser, vapor/liquid separator, vacuum pump and a makeup water tank. The vacuum pump is connected to the loop containing the solution which contains 10wt% sugar and it controls the pressure in the separator. Heat exchanger transfer heat between the two fluids. Steam is used to heat the sugar in the vapor/liquid separator to its boiling point. Then the vapor is transferred to the condenser which

Evaporation

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uses cooling water to bring the vapor to its liquid state. Condensed vapor was then collected in a volumetric flask to calculate its capacity and economy. Vacuum pressure was the only parameter that was changed in the system and changed the boiling point of the solution. A make-up tank was connected to the liquid/vapor separator to keep the volume of the sugar solution constant by adding water to sugar solution when the level falls. This was to make sure the concentration of the sugar solution remains the same throughout the process. The steam trap is also used to condense the remaining steam. The condensed vapor was collected to calculate the economy. The solution was prepared by mixing 280 g of sugar with 2.8 L of distilled water and then was added to the liquid/vapor separator, Also distilled water was added to the makeup tank. Cooling water to steam trap and steam to heat exchanger were turned on. Vacuum pump was turned on and set at 10 inHg for 15 minutes. The volume of the steam condensate and collected vapor was measured after the system reached the steady state. After 15 minutes this process was repeated for 15 in Hg and again after another 15 minutes was repeated for 20 in Hg. At the inlet and outlet of the steam trap the temperature of the cooling water and its flow rate were measured for each one of the three trials which was used to calculate the amount of steam condensed in the heat exchanger [5].

Commented [DL10]: Keep your citations consistent.

Evaporation is preferred in industry over other alternative methods because it requires less energy and results in lower cost therefore it is widely used in industries such as agriculture, waste management, water treatment natural oil, etc. it is used to clean sea water, in oil field evaporation is used to separate water from crude oil. It is used in food industry to condensate some liquid foods such as noodles and milk. Also used in pharmaceutical companies to improve the quality of the drugs by removing the water moisture [6].

Commented [DL11]: Good.

3. Theory

Evaporation occurs when the given element absorbs enough energy to transition from a liquid phase into a gas phase. When this transition is occurring, the element will stay at constant temperature. This constant temperature is known as the boiling point. The boiling point temperature can most definitely be changed by having fluctuations in pressure. There is a proportional relationship between the two variables of pressure and temperature (boiling point). As the pressure increases in a system, temperature of the boiling point also increases and while the pressure decreases in a system, the temperature of the boiling point will also decrease.

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Include the overall heat transfer equation: $q = UAT$.

Due to this experiment is calling for using a pump, the pump will behave as variable to take away the pressure. The reduction of pressure will conclude in the solvent and having a lower boiling point. The energy required now is much more insignificant for the solvent to evaporate because of the lower boiling point. Thermal energy entering will stay constant because of the amount of solvent that is going to be evaporated. This will straightforwardly influence the evaporation efficiency.

Commented [DL13]: Good.

Steam condensate volumes were recorded to analyze the capacity and economy. Capacity is the ratio of the total amount of distillate times the density of water per time. Its formula goes as follows:

$$Capacity = \frac{V_{dist} \cdot \rho_w}{t}$$

Where:

- V_{dist} = volume of Distillate Collected [ml]
- ρ_w = Density of water [$\frac{kg}{ml}$]

- $t = \text{Time [hr]}$

The flowrate of cooling water and its respected temperature both at the inlet and outlet, the energy from the stream trap to the heat exchanger, and the total energy from the heat exchanger were documented. It's given formula goes as follows:

$$q_{lb} = \dot{m}_{cw} * C_p * \Delta T$$

Where:

- $C_p = \text{Specific Heat Capacity of water } [\frac{kJ}{kg} * K]$
- $\dot{m}_{cw} = \text{Mass flow rate of Steam } [\frac{kg}{hr}]$
- $\Delta T = \text{Temperature Change of colling water in the steam Trap}[K]$

The overall heat balance for both the total system and the heat exchanger system was calculated. The formula for the two indices goes as follows:

Overall Heat Balance:

$$q_{total} = \dot{m}_s * C_p * \Delta T + \dot{m}_s \Delta H_{vap}$$

Where:

- $C_p = \text{Specific Heat Capacity of water } [\frac{kJ}{kg} * K]$
- $\dot{m}_s = \text{Mass flow rate of Steam } [\frac{kg}{hr}]$
- $\Delta T = \text{Temperature Change of colling water in the steam Trap}[K]$
- $\Delta H_{vap} = \text{Enthalpy of Vaporization of water}$

Commented [DL14]: Units.

Heat Exchanger Balance:

$$q_x = q_t - q_{st}$$

Where:

- q_x = Heat Transfer Rate of Heat Exchanger [KJ/hr]
- q_t = Total Heat Transfer Rate over Exchanger and Steam Trap [kJ/hr]
- q_{st} = Heat Transfer Rate of Steam Trap $\left[\frac{kJ}{hr}\right]$

Finally, the total economy is calculated. It is the ratio of the mass flow rate of the condensed vapor per mass flow rate of steam. Its respected formula goes as follows:

$$Economy = \frac{\dot{m}_v}{\dot{m}_s}$$

Where:

- $\dot{m}_v$ = Mass flow rate of Condensed Vapor $\left[\frac{kg}{hr}\right]$
- $\dot{m}_s$ = Mass flow rate of Steam $\left[\frac{kg}{hr}\right]$

4. Results

A parallel analysis of capacity and economy of a small-scale evaporation experiment constituting of a 10wt% sucrose solution over three different sets of vacuum pressures was done. Moreover, calculation of energy balance on the system was performed to make sense of the underlying parameters of heat transfer in the system. Three trials were performed at a vacuum pressure of respectively 10,15 and 20 inches of Hg. The condensates of both the steam and vapor streams were collected over a 15-minute period and the final collected volumes were recorded used for analysis. A constant cooling water flow rate and temperature differences across certain parts of the system were recorded for each trial and used in calculations of evaporator capacity and economy. The raw data is provided in appendix I and the figures are shown below.

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Include the Vac. Vs. HX Energy and the Vac. Vs. Differential T figures.

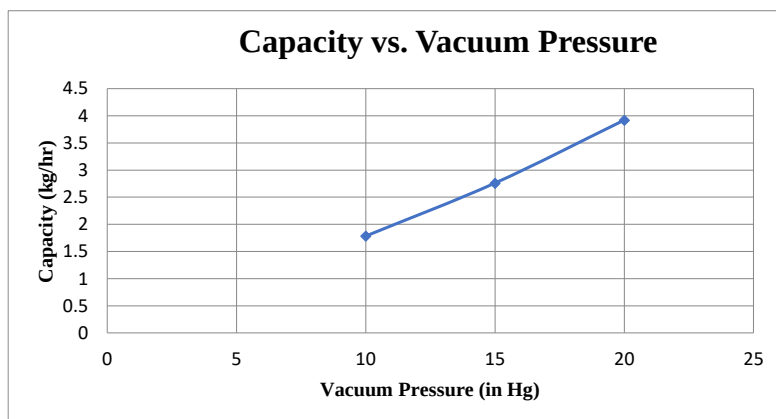


Figure 1: Correlation between Capacity and Vacuum pressure

Figure 1 displays the correlation between Capacity of the evaporation process over three vacuum pressures at 10, 15 and 20 in Hg. The capacity at such pressures are respectively 1.784,

2.76, and 3.92 kg/h. The figure clearly shows that the capacity of the evaporator linearly increases as the vacuum pressure increase over the range of 10 and 20.

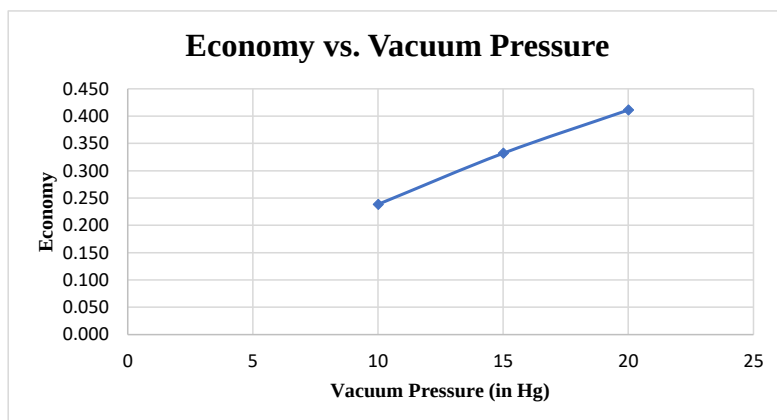


Figure 2: Correlation between Economy and Vacuum Pressure

Figure 2 displays the correlation between economy of the evaporation process over three vacuum pressures at 10, 15 and 20 in Hg. The economy at such pressures are respectively 0.238, 0.332 and 0.411 kg steam used / kg product. The figure clearly indicates that the economy of the evaporator linearly increases as the vacuum pressure increase over the range of 10 and 20. Such result illustrates the advantages of using a vacuum system in tandem with an evaporation unit to more efficiently purify a multicomponent mixture. However, it is uncertain to generalize the trend for an entire process because the experiment was conducted within a range of 10 to 20 in Hg pressure values.

The boiling point of the sugar water solution decreased with increasing pressure contrary to the increase in economy and capacity with increased pressure. Such observation is consistent

with steam table data for pure water. Moreover, the boiling point of the sugar water solution was observed to be slightly lower than that of pure water at the pressures of 10, 15 and 20 in Hg. The reason might be that the boiling point of a mixture is inversely proportional to the pressure of the mixture. Such result is inconsistent with Raoult's Law, which suggests that a solution of a non-volatile solid dissolved in water should display an increase in boiling point. The exact cause of such discrepancy is unknown. However, a trial with pure water in this system could hypothetically display the expected behavior. This could demonstrate a systemic flaw in the measurement method, such as the location of the thermocouple being inadequate for measuring the actual boiling point of the solution.

Commented [DL16]: The solution is neither dilute nor ideal.

Commented [DL17]: It doesn't affect the results a lot.

Commented [DL18]: Good discussion.

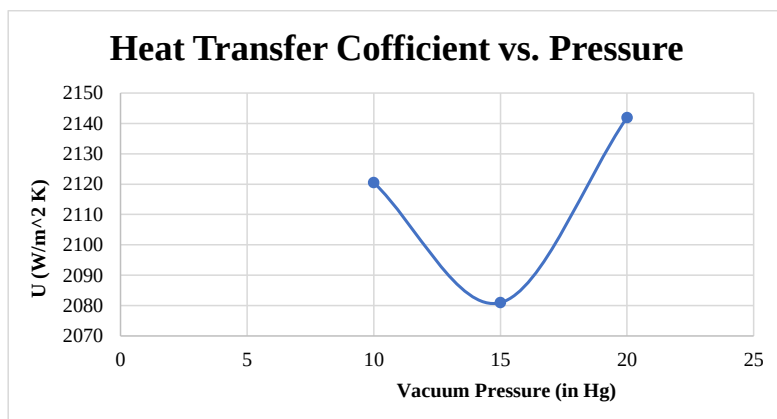


Figure 3: Correlation between Heat Transfer Coefficient and Vacuum Pressure

Figure 3 displays the correlation between heat transfer coefficient of the evaporation process over three vacuum pressures at 10, 15 and 20 in Hg. Any increase in steam pressure provides energy in the form of work, and such energy would be expressed as an increase in the steam's enthalpy of vaporization. Since the calculations for steam used do not take the pressure drop into

account, its exact effect on the heat exchange is unknown. However, the effect of the increase in apparent available steam is essentially accounted for in the overall economy of the evaporation unit and its specific contribution was ignored for the purposes of the experiment.

The overall heat transfer coefficient U was calculated for the heat exchanger itself using the design equation for heat exchangers (Appendix II). Such values ranged from 2121 W/m²·K (10 inHg trial) to 2142 W/m²·K (20 inHg trial) [Figure 3]. The average across all three trials was 2115 W/m²·K. The heat transfer coefficient should be relatively constant across each trial, but the experimental results show changing values for U . Such discrepancy is likely due to the uncertainty in measurements (Appendix III).

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5. Discussion

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The applied vacuum pressure influenced the economy and capacity of the process. The reduced pressure increased the amount of water evaporated from the water-sugar solution. The direct effect of the vacuum pressure was to reduce the boiling point of the solution, such reduction lead to several other effects. The first effect was that make-up water needed to be heated less upon entering the separation vessel to reach its boiling point. This heat duty reduction allowed more of the heat transferred by the steam to be utilized in vaporizing the solution. Another effect was that the sugar-water solution was at a lower temperature reducing the temperature difference between the solution and ambient temperature. This reduced temperature difference resulted in less heat loss to the surroundings, which also allowed more heat transferred by the steam to vaporize the water from solution.

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Commented [DL22]: Good discussion.

The results showed a linear dependency of pressure on the economy of the evaporator as briefly mentioned above. The economy increased proportionally with increasing pressure. The economy values obtained were 0.411, 0.332, and 0.238 kg steam used per kg water evaporated for 20, 15 and 10 in. Hg pressures. Economy is a ratio of steam used per water in the solution evaporated. Ideally, in multi-stage evaporators, the economy is desired to be greater than one (1) to increase efficiency and decrease energy losses. However, in a single-effect evaporator, unity is difficult to achieve due to thermodynamic irreversibility; therefore, values are usually significantly less than unity. In industry, good economy values in a single-effect evaporation are between 0.769–1.000 kg water evaporated per kg steam used for the evaporation [5]. A low economy value such as 0.238 is completely undesirable unless the heat used is at the end of the process, where the heat would

otherwise be unusable and wasted. In a situation like that, if the cost of operation is negligible, the heat can be used to reheat another stream.

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Data analysis showed that the capacity also increased with pressure. The values obtained were of 3.92, 2.76, and 1.78 kg/hr of water for 10, 15 and 20 in. Hg pressures. The higher the capacity, the more water was evaporated from the solution. In industry, maximizing capacity could potentially lead to more profit; however, further analysis needs to be conducted to determine how an increase in pressure affects the associated energy losses. In the experiment, energy losses by steam in the heat exchanger were calculated to 26170, 23547, and 22045 J/s for 20, 15 and 10 in. Hg pressures. Even though the economy of the evaporator increased with increasing pressure, the associated energy loss in the heat exchanger also increase.

Commented [DL24]: You should include this in the previous section.

6. References

1. "Properties of Water." *Wikipedia*, Wikimedia Foundation, 3 Feb. 2020, en.wikipedia.org/wiki/Properties_of_water.
2. J. Moran, H. Shapiro. "Fundamentals of Engineering Thermodynamics." John Wiley and Sons, NJ: 2011.
3. Bird, Stewart, and Lightfoot. "Transport Phenomena." John Wiley and Sons, NJ: 2007.
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6. Thermal Kinetics. "Common Uses and Applications for Evaporators." *Thermal Kinetics*, Thermal Kinetics, 23 Aug. 2019, thermalkinetics.net/common-uses-applications-evaporators.

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[2], [3] and [4] are not found in the report.

Commented [DL26]: Pages.

Commented [DL27]: Try to follow some of the citation standards.

7. Appendix I: Data Tabulation/Graphs

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Specific Heat Capacity (KJ/kg*K) ^[1]	4.187
Enthalpy of Condensing Water (KJ/kg) ^[1]	2260
Density Water (kg/ml) ^[1]	0.001

Table 1: Vacuum pressures, Temperatures, Flow rates and Heat balances for Evaporation Lab

Trial	3	2	1
Vacuum Pressure (in Hg)	20	15	10
Water Vapor Temp. (HE Outlet) (°C)	69	78	87
Water Vapor Temp. (Condenser inlet) (°C)	68	76	88
Condensed Water Vapor Temp. (Outlet) (°C)	20	18	16
Steam Temp. (Out from HE, inlet to ST) (°C)	100	102	103
Cooling Water Temp. (Outlet from ST) (°C)	26	29	34
Cooling Water Temp. (Inlet from ST) (°C)	7	7	8
Condensed Steam Temp. (out from ST) (°C)	32	36	45
Inlet Steam Temp. (HE Inlet)(°C)	114	113	114
Cooling water Flow Rate(0.536 gal/min)	0.188	0.188	0.188
Heat Exchanger Pressure (kPa)	19	24	28
Distillate Volume (mL)	980	690	446
Steam Volume (mL)	2571	2321	2202
Elapsed Time (min)	15	15	15
Elapsed Time (hr)	0.25	0.25	0.25
Makeup Water Hight Difference (cm)	9.75	4.1	3.3
Makeup Water Used (mL)	1387.76	583.57	469.70
Capacity			
Distillate Flow Rate (kg/h)	3.92	2.76	1.784
Steam Flow Rate (kg/h)	10.284	9.284	8.808
Log Mean Temp. Difference	58.30	51.69	44.76
Rate of Heat Transfer in ST (kJ/h)	3390	3925	4638
Rate of Heat Transfer from ST to HE (kJ/h)	26170	23547	22045
Rate of Heat Transfer in HE (kJ/h)	22780	19623	17407
Rate of steam used (kg/h)	9.532	8.313	7.480
Economy	0.411	0.332	0.238
Heat Transfer Coefficient (W/m ² K)	2142	2081	2121

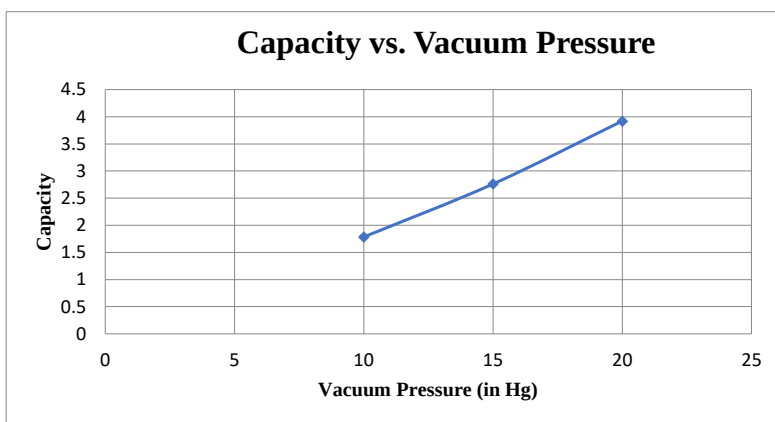


Figure 4: Correlation between Capacity and Vacuum pressure

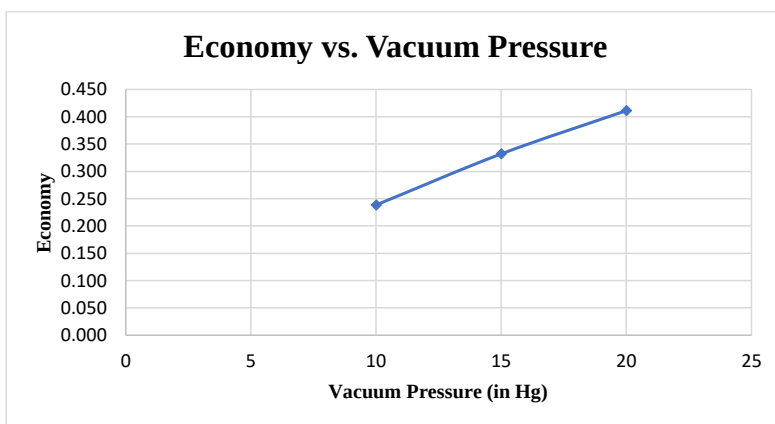


Figure 5: Correlation between Economy and Vacuum Pressure

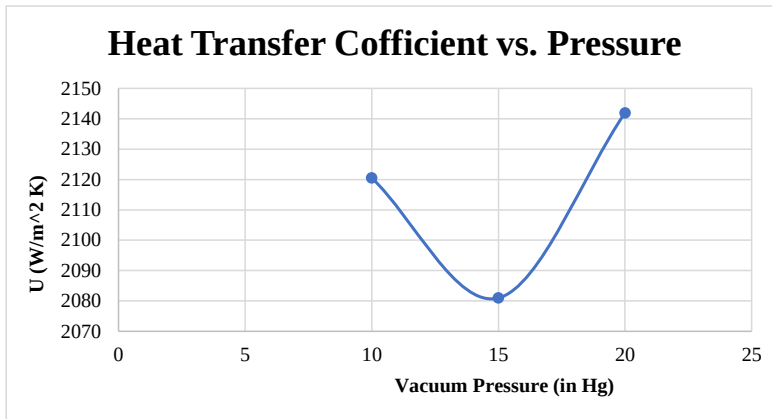


Figure 6: Correlation between Heat Transfer Coefficient and Vacuum Pressure

8. Appendix II: Error Analysis

During this experiment number of technical and non-technical error may have caused unintended errors. Accurate measurement of sugar solution, pouring in water in the distillate vessel without spill over, steam and distillate measurement accurately, can be mention as error causing reasons.

- 1. Instrument Error:** From get go, and one of the pressure gauges is not functional. The bottom pressure controller is not calibrated for setting up or maintaining pressure throughout the experiment. The temperature reader fluctuates while taking readings up to plus or minus 2 degree. Hence it is advisable to take time for accurate reading.
- 2. Sampling Errors:** In this experiment, all the 9 samples were supposed to be taken during a steady state. But in the attempt to maintain the water level in the distillate level, the water added may disturb the steady state. Though 15 minutes is required between the experiments, sometimes it may require an extended time and even to retake the sample. This also may be responsible the temperature fluctuation that is observed.
- 3. Human Errors:** The highest probability for human error could be the level reading in graduated cylinder and time measurement. In this experiment some of the fluids flowed into the unintended tube, which lead to repeat the whole task. And [also](#), the back flow from closing and opening the distillate may create a flow accumulation in the tubes and create error reading.

Commented [DL29]: 8/10

Quantitative analysis is needed. What if the solution level varies ~5%? Try to make error bars according to this.

9. Appendix III: Sample Calculations

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Specify the calculation you presented.

Liquid Balance for cooling water

$$q_{lb} = \dot{m}_{cw} * C_p * \Delta T$$

$$C_p = \text{Specific Heat Capacity of water} \left[\frac{kJ}{kg} * K \right]$$

$$\dot{m}_{cw} = \text{Mass flow rate of Steam} \left[\frac{kg}{hr} \right]$$

$$\Delta T = \text{Temperature Change of cooling water in the steam Trap} [K]$$

$$q_{lb} = 0.188 \left(\frac{gal}{min} \right) * 3.7854 \left(\frac{l}{gal} \right) * 60 \left(\frac{min}{hr} \right) * 1 * \left(\frac{kg}{l} \right) * 4.187 \left(\frac{kJ}{kg * K} \right) * (26 - 7)$$

$$q_{lb} = 3396.8 \left(\frac{kJ}{hr} \right)$$

Over all Q balance

$$q_{total} = \dot{m}_s * C_p * \Delta T + \dot{m}_s \Delta H_{vap}$$

$$C_p = \text{Specific Heat Capacity of water} \left[\frac{kJ}{kg} * K \right]$$

$$\dot{m}_s = \text{Mass flow rate of Steam} \left[\frac{kg}{hr} \right]$$

$$\Delta T = \text{Temperature Change of cooling water in the steam Trap} [K]$$

$$\Delta H_{vap} = \text{Enthalpy of Vaporization of water}$$

$$\begin{aligned} q_{total} &= 10.284 \left(\frac{kg}{hr} \right) * 4.187 \left(\frac{kJ}{kg * K} \right) * (26 - 7) + 10.284 \left(\frac{kg}{hr} \right) * 2260 \left(\frac{kJ}{kg} \right) \\ &= 24059.9 \left(\frac{kJ}{hr} \right) \end{aligned}$$

Heat Exchanger Balance

$$q_x = q_t - q_{st}$$

$$q_x = \text{Heat Transfer Rate of Heat Exchanger} [KJ/hr]$$

Unit Operations CHE 382 Group 1

Thursday, AM section

Spring 02/5/2020

Evaporation

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q_t = Total Heat Transfer Rate over Exchanger and Steam Trap [kJ/hr]

q_{st} = Heat Transfer Rate of Steam Trap [kJ/hr]

C_p - Specific Heat Capacity of Water [kJ/(kg * K)]

ΔT - Temperature Change Over the Heat Exchanger [K]

ΔH_{vap} = Enthalpy of Vaporization of water

$$= 24059.9 \left(\frac{kJ}{hr} \right) - 3396.8 \left(\frac{kJ}{hr} \right)$$

$$q_x = 20663.1 \left(\frac{kJ}{hr} \right)$$

$$\dot{m}_s = \frac{q_x}{C_p * \Delta T + \Delta H_{vap}}$$

$$\dot{m}_s = \frac{20663.1 \left(\frac{kJ}{hr} \right)}{4.187 \left(\frac{kJ}{kg * K} \right) * (100 - 32) + 2260 \left(\frac{kJ}{kg} \right)}$$

$$\dot{m}_s = 9.408 \frac{kg}{hr}$$

Heat Transfer Coefficient

$$Q = U * A * \Delta T_{LM}$$

$$\Delta T_{LM} = \frac{((114 - 100) - (69 - 26))}{\ln \left(\frac{(114 - 100)}{(69 - 26)} \right)}$$

$$T_{LM} = 12.47 \text{ K}$$

Q - Rate of Heat Transfer [J/s]

U - Heat Transfer Coefficient [W/(m² * K)]

A - Heat Exchanger Area [m²]

ΔT_{LM} - Log Mean Temperature Difference [K]

Evaporation

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$$U = \frac{20663.1 \left(\frac{kJ}{hr} \right) * \left(\frac{1hr}{3600s} \right) * \left(\frac{1000j}{kJ} \right)}{0.0506 m^2 * 12 K}$$

$$= 9452.8 \frac{W}{m^2 * K}$$

Capacity is the amount of product (Condensate) collected per time (kg/hr)@ 20 in Hg

$$Capacity = \frac{V_{dist} * \rho_w}{t}$$

V_{dist} = volume of Distillate Collected [ml]

ρ_w = Density of water $\left[\frac{kg}{ml} \right]$

t = Time [hr]

$$= \frac{980 ml * 0.001 kg/ml}{0.25 hr}$$

$$Capacity = 3.92 \left(\frac{kg}{hr} \right)$$

Economy: is the ratio of (kg of product/ kg of steam used) @ 20 in Hg

$$Economy = \frac{\dot{m}_v}{\dot{m}_s}$$

$\dot{m}_v$ = Mass flow rate of Condensed Vapor $\left[\frac{kg}{hr} \right]$

$\dot{m}_s$ = Mass flow rate of Steam $\left[\frac{kg}{hr} \right]$

$$= \frac{3.92}{9.532}$$

$$Economy = 0.411$$

10. Appendix IV: Individual Team Contributions

NAME: *Mahamat Abdraman*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Present in both sessions
Pre-Lab Editing	1	Group work
Experimental Objective (Prelab)	1	Wrote section
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	2	Edited 2 nd draft
Executive Summary (final lab)	1	Wrote first half draft
Introduction (Final lab)	1	Wrote first half draft
Theory (Final Lab)	0	
Results (final lab)	8	Draft
Discussion (final lab)	2	Group of two work
References (final and/or prelab)	0.5	
Data Tabulation/Graphs (final)	3	Group of two work
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	3	First draft
Total	30.5	

NAME: Gewargis Garmooos

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Did the experiment
Pre-Lab Editing	.5	Editing
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	3	Wrote the procedure
Project Safety Evaluation (prelab)	.5	Reviewed the safety
Final Lab Editing	.5	Editing
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	3	Wrote the theory
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	15.5	

NAME: *Sinan Masoud*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Present both days
Pre-Lab Editing	1	Group editing
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	1	Wrote section
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0	
Executive Summary (final lab)	1	Wrote second half
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	
Total	11	

NAME: *Marvl Odeesh*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Working on the lab
Pre-Lab Editing	.5	Editing the lab
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	2	
Final Lab Editing	2	Editing
Executive Summary (final lab)		
Introduction (Final lab)	3	Wrote introduction
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)	.5	
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	16	

NAME: *Yohannes Teweldeberhan*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	Present Both days
Pre-Lab Editing	1	Formatting
Experimental Objective (Prelab)		
Apparatus (prelab)	3	
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	0.5	
Final Lab Editing	3	
Executive Summary (final lab)	1	
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	5	Writing
Sample Calculations (final Lab)	5	Calculation and typing
Power Point Presentation		
Total	23	

Evaporation

University of Illinois

Signature of Team Member	Date
Marvel Odeesh	2/5/2020
Gewargis Garmooos	2/5/2020
Mahamat Abdraman	2/5/2020
Sinan Masoud	2/5/2020
2/5/2020	2/5/2020